Chapter 3, Section 8

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1. Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{F}$ be a sheaf of Abelian groups on X, and assume that $\mathrm{R}^i f_*(\mathcal{F}) = 0$ for all $i > 0$. Show that there are natural isomorphisms, for each $i \geq 0$,

$$\mathrm{H}^i(X, \mathcal{F}) \cong \mathrm{H}^i(Y, f_*\mathcal{F}).$$

Proof. Let $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution of $\mathcal{F}$. By hypothesis, $0 \to f_*\mathcal{F} \to f_*\mathcal{I}^\bullet$ is also exact. The pushforward of an injective object in $\mathfrak{Ab}(X)$ is also injective in $\mathfrak{Ab}(Y)$ because f_* is a right adjoint functor, so $0 \to f_*\mathcal{F} \to f_*\mathcal{I}^\bullet$ is an injective resolution of $f_*\mathcal{F}$ in $\mathfrak{Ab}(Y)$. Suppose $\mathcal{G} \to \mathcal{G}'$ is a monomorphism of sheaves on Y, and $\mathcal{G} \to f_*\mathcal{I}$ is any morphism of sheaves. By adjunction, there is a morphism $f^{-1}\mathcal{G} \to \mathcal{I}$, and because f^{-1} is an exact functor, $f^{-1}\mathcal{G} \to f^{-1}\mathcal{G}'$ is also a monomorphism (note that in general, f^* is only right exact). By injectivity of $\mathcal{I}$, there is an extension of $f^{-1}\mathcal{G} \to \mathcal{I}$ making the following diagram commute

$$\begin{array}{ccc} & & \mathcal{I} \\ & \nearrow & \uparrow \\ f^{-1}\mathcal{G} & \longrightarrow & f^{-1}\mathcal{G}' \end{array} .$$

Hence, by adjunction again, there is an extension of $\mathcal{G} \to f_*\mathcal{I}$ making the following diagram commute

$$\begin{array}{ccc} & & f_*\mathcal{I} \\ & \nearrow & \uparrow \\ \mathcal{G} & \longrightarrow & \mathcal{G}' \end{array} .$$

Since $\mathcal{F}$ and $f_*\mathcal{F}$ have same global sections for any $\mathcal{F}$, there are natural identificiations

$$\begin{array}{ccccc} 0 & \longrightarrow & \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(X, f_*\mathcal{I}^\bullet) \\ & & \| & & \| \\ 0 & \longrightarrow & \Gamma(Y, f_*\mathcal{F}) & \longrightarrow & \Gamma(Y, f_*\mathcal{I}^\bullet) \end{array} .$$

Taking cohomology gives the desired natural isomorphisms for all $i \geq 0$. □

2. Let $f : X \to Y$ be an affine morphism of schemes (II, Ex. 5.17) with X Noetherian, and let $\mathcal{F}$ be a quasi-coherent sheaf on X. Show that the hypotheses of (Ex. 8.1) are satisfies, and hence that $\mathrm{H}^i(X, \mathcal{F}) \cong \mathrm{H}^i(Y, f_*\mathcal{F})$ for each $i \geq 0$.

Proof. Let V be an open affine subset of Y. By (8.2) and (8.5), we have

$$\mathrm{R}^i f_*(\mathcal{F})\big|_V \cong \widetilde{\mathrm{H}^i(f^{-1}V, \mathcal{F}\big|_{f^{-1}(V)})},$$

which is zero because $f^{-1}(V)$ is affine (3.7). Hence, $\mathrm{R}^i f_*(\mathcal{F}) = 0$, and the result follows by the previous exercise. □

3. Let $f : X \to Y$ be a morphism of ringed spaces, let $\mathcal{F}$ be an $\mathcal{O}_X$-module, and let $\mathcal{E}$ be a locally free $\mathcal{O}_Y$-module of finite rank. Prove the *projection formula* (cf. (II, Ex. 5.1))

$$\mathrm{R}^i f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong \mathrm{R}^i f_*(\mathcal{F}) \otimes \mathcal{E}.$$

Proof. The pullback of a locally free sheaf of finite rank is also locally free of finite rank. Localization is an exact monoidal functor from the category of $\mathcal{O}_Y$-modules to the category of Abelian groups, so by checking exactness at stalks, tensoring by $\mathcal{E}$ is an exact functor, and it also has a left adjoint, where for any $\mathcal{O}_Y$-modules $\mathcal{M}$ and $\mathcal{N}$ (II, Ex. 5.1)

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M} \otimes \mathcal{E}^\vee, \mathcal{N}) &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{E}^\vee, \mathcal{N})) \\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathcal{N} \otimes (\mathcal{E}^\vee)^\vee) \\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathcal{N} \otimes \mathcal{E}).\end{aligned}$$

Hence, the result follows from the $i = 0$ case, (1.3A), and Lemma 1.

It remains to show

$$f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong f_*(\mathcal{F}) \otimes \mathcal{E}.$$

We first show there is a natural map

$$\phi : f_*(\mathcal{F}) \otimes \mathcal{E} \to f_*(\mathcal{F} \otimes f^*\mathcal{E}).$$

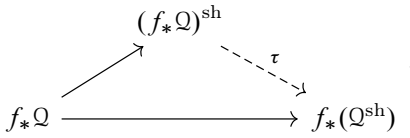
Let $\mathcal{P}$ be the presheaf on Y defined for open subsets $V \subseteq Y$ by

$$V \mapsto f_*(\mathcal{F})(V) \otimes \mathcal{E}(V)$$

so that $\mathcal{P}^{\mathrm{sh}} \cong f_*(\mathcal{F}) \otimes \mathcal{E}$, and let $\mathcal{Q}$ be the presheaf on X defined for open subsets $U \subseteq X$ by

$$U \mapsto \mathcal{F}(U) \otimes f^*\mathcal{E}(U)$$

so that $\mathcal{Q}^{\mathrm{sh}} \cong \mathcal{F} \otimes f^*\mathcal{E}$. Sheafification and pushforward do not commute; nonetheless there is a natural morphism $(f_*\mathcal{Q})^{\mathrm{sh}} \to f_*(\mathcal{Q}^{\mathrm{sh}})$. In particular, consider the natural morphism $\mathcal{Q} \to \mathcal{Q}^{\mathrm{sh}}$. Taking the pushforward gives $f_*\mathcal{Q} \to f_*(\mathcal{Q}^{\mathrm{sh}})$. Because $f_*(\mathcal{Q}^{\mathrm{sh}})$ is already a sheaf, by the universal property of sheafification (II, 1.2), there is unique map $\tau : (f_*\mathcal{Q})^{\mathrm{sh}} \to f_*(\mathcal{Q}^{\mathrm{sh}})$ fitting into the commutative diagram



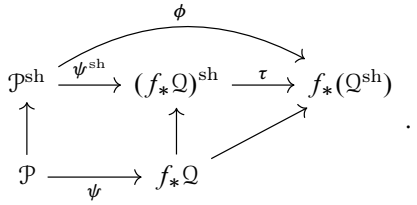
Thus, we define a morphism of presheaves $\psi : \mathcal{P} \to f_*\mathcal{Q}$, which induces a morphism of sheaves $\psi^{\mathrm{sh}} : \mathcal{P}^{\mathrm{sh}} \to (f_*\mathcal{Q})^{\mathrm{sh}}$, and composing with τ gives our desired morphism

$$\phi = \tau \circ \psi^{\mathrm{sh}} : \mathcal{P}^{\mathrm{sh}} \to (f_*\mathcal{Q})^{\mathrm{sh}} \to f_*(\mathcal{Q}^{\mathrm{sh}}).$$

The pushforward for the presheaf $\mathcal{Q}$ distributes over the tensor product, that is

$$\Gamma(V, f_*\mathcal{Q}) = f_*(\mathcal{F})(V) \otimes f_*f^*(\mathcal{E})(V).$$

There is a natural morphism of sheaves $\iota : \mathcal{E} \to f_*f^*(\mathcal{E})$, so take $\psi = \mathrm{id}_{f_*\mathcal{F}} \otimes \iota$ to be our morphism of presheaves



Now that we defined a morphism, we remark that the question is local on Y. In particular, let y be a point in Y, and let V be an open subset of Y such that $\mathcal{E}$ is free of finite rank. Then $f^*\mathcal{E}$ is free of finite rank on $U = f^{-1}(V)$. Consider the commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{\bar{f}} & V \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ X & \xrightarrow{f} & Y \end{array}.$$

Because i and j are open immersions, there is an isomorphism of functors

$$j^* f_* \cong \bar{f}_* i^*, \quad i^* f^* \cong \bar{f}^* j^*.$$

Pullback distributes over tensor products, so we have

$$\begin{aligned} j^* f_*(\mathcal{F} \otimes f^*\mathcal{E}) &\cong \bar{f}_* i^*(\mathcal{F} \otimes f^*\mathcal{E}) \\ &\cong \bar{f}_*(i^*\mathcal{F} \otimes i^* f^*\mathcal{E}) \\ &\cong \bar{f}_*(i^*\mathcal{F} \otimes \bar{f}^* j^*\mathcal{E}), \\ j^*(f_*(\mathcal{F}) \otimes \mathcal{E}) &\cong j^* f_*(\mathcal{F}) \otimes j^*\mathcal{E} \\ &\cong \bar{f}_*(i^*\mathcal{F}) \otimes j^*\mathcal{E}. \end{aligned}$$

Open immersions preserve stalks, so we can assume $\mathcal{E}$ to be a free $\mathcal{O}_X$-module of finite rank n. The pullback of a free sheaf of finite rank is also free of finite rank, so we have the isomorphisms

$$\begin{aligned} f_*(\mathcal{F} \otimes f^*\mathcal{E}) &\cong f_*(\mathcal{F} \otimes \mathcal{O}_X^{\oplus n}) \\ &\cong f_*(\mathcal{F}^{\oplus n}) \\ &\cong f_*(\mathcal{F})^{\oplus n} \\ &\cong f_*(\mathcal{F}) \otimes \mathcal{O}_Y^{\oplus n} \\ &\cong f_*(\mathcal{F}) \otimes \mathcal{E}. \end{aligned}$$

The third isomorphism holds because f_* is right adjoint, and right adjoint functors preserve limits, and a direct sum of finite number of terms can be realized as a limit, so the pushforward and finite direct sums commute. □

Lemma 0.1. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be Abelian categories, and assume $\mathcal{A}$ and $\mathcal{B}$ have enough injectives. Let $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{C}$ be functors such that F has a left adjoint (or more generally preserves injectives), and either*

(a) F is left exact, and G is exact; or

(b) F is exact, and G is left exact.

In each case, we have the following natural isomorphisms

(a) $\mathrm{R}^i(G \circ F) \cong G \circ \mathrm{R}^i(F)$;

(b) $\mathrm{R}^i(G \circ F) \cong \mathrm{R}^i(G) \circ F$.

Proof. The proof for (b) is similar to the proof of Exercise 1, so we prove (a). Let A be an object in $\mathcal{A}$ and choose an injective resolution

$$0 \to A \to I^\bullet,$$

where one has

$$\mathrm{R}^i F(A) = h^i(FI^\bullet).$$

Applying $G \circ F$ gives a complex in $\mathcal{C}$

$$0 \to (G \circ F)A \to (G \circ F)I^\bullet,$$

and we want to show

$$h^i((G \circ F)I^\bullet) \cong G \circ h^i(FI^\bullet).$$

But this follows from exact functors commuting with cohomology.

□